

2. If you want to travel to The UK, but you can only speak Chinese. In order to talk with the natives, you may need a translator. Now suppose that you have a ChineseSpeaker class that implements the SpeakChinese interface and a EnglishSpeaker class that implements the SpeakEnglish interface. Design a LanguageAdapter class that using the adapter pattern to solve this problem. Write a Test class to test the result.

ChineseSpeaker

```
package experiment8_2;

class ChineseSpeaker implements SpeakChinese{

    @Override

    public void speakChinese() {

        System.out.println("Speak in Chinese: 你好");

    }

}
```

EnglishSpeaker

```
package experiment8_2;

class EnglishSpeaker implements SpeakEnglish{

    @Override

    public void speakEnglish() {

        System.out.println("Speak in English: Hello");

    }

}
```

LanguageAdapter

```

package experiment8_2;

class LanguageAdapter implements SpeakEnglish{
    private SpeakChinese chineseSpeaker;

    public LanguageAdapter(SpeakChinese chineseSpeaker){
        this.chineseSpeaker = chineseSpeaker;
    }

    @Override
    public void speakEnglish() {
        chineseSpeaker.speakChinese();
        System.out.println("Translating Chinese to English
by adapter: Hello");
    }
}

```

SpeakChinese

```

package experiment8_2;

interface SpeakChinese {
    void speakChinese();
}

```

SpeakEnglish

```
package experiment8_2;

interface SpeakEnglish {
    void speakEnglish();
}
```

Test

```
package experiment8_2;

public class Test {
    public static void main(String[] args) {
        SpeakChinese chineseSpeaker = new
ChineseSpeaker();

        SpeakEnglish adapter = new
LanguageAdapter(chineseSpeaker);

        adapter.speakEnglish();
    }
}
```

Output

```
D:\JDK\bin\java.exe "-javaagent:D:\idea\IntelliJ IDEA
```

```
Speak in Chinese: 你好
```

```
Translating Chinese to English by adapter: Hello
```

```
Process finished with exit code 0
```